

Roll No.

TEE-504

**B. TECH. (EE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

ELECTRICAL MACHINE DESIGN

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *one* sub-questions among (a) and (b) in each main question.

(iii) Each question carries 10 marks.

1. (a) Differentiate between a single phase core type transformer and shell type transformer. (CO1)

OR

- (b) Discuss window space factor in transformer design. (CO1)

2. (a) What is meant by specific electric and specific magnetic loading in a dc machine ? (CO2, CO3)

OR

- (b) Explain the procedure for separation of D and L from DL product while designing induction motors. (CO2, CO3)

P. T. O.

(2)

3. (a) Derive the relation for KVA rating of core type single phase transformer. (CO1)

OR

- (b) Determine the number of poles, armature diameter and core length for the preliminary design of a 500 kW, 400V, 600 r.p.m. dc shunt generator assuming an average flux density in the air gap of 0.7 T and specific electric loading of 38400 ampere-conductors per metre. Assume core length/pole arc = 1.1. (CO1)

4. (a) What are the factors to be considered for design of flux density of the transformer? (CO2)

OR

- (b) Enumerate the importance of No-load current of transformer with suitable diagram. (CO2)

5. (a) A design is required for a 50 kW, 4 pole, 600 r.p.m., and 220V dc shunt generator. The average flux density in the air gap and specific electric loading are respectively 0.57T and 30000 ampere-conductors per metre. Calculate suitable dimensions of armature core to lead to a square pole face. (CO3)

OR

- (b) What are the factors on which output power of dc machine depends? (CO3)